

eVTOL Patent Figures and DINOv2 Embeddings: Progress Report

Prepared for: Phase Supervisor

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Material: 675 aircraft from 583 patents, 1567 whole-aircraft figures (1 639-patent dataset, all five batches)

Model: facebook/dinov2-large (frozen), input 518 px, layers {18, 22, 24} × pooling {cls, mean_patch}

Successor of `Supervisor_Report_Taxonomy_Structure_Analysis_summary.md` (Batches 01 + 05, 264 images, July labels). Same layout and tests, now on the whole labelled dataset. Generated by `embedding_evaluation/scripts/build_embedding_analysis.py`; every number is computed from the files listed at the end.

Summary

- **Images.** The probes need one vector per aircraft, and most aircraft are drawn more than once. Four rules choose that vector for every one of the 675 aircraft that has a G1 code. Three pick a single figure (by view, by flight state, or the labeller's main figure). The fourth averages one figure per view.
- **How much the rules differ.** View-first and state-first pick the same figure for 96.6 % of aircraft. The labeller's main figure differs from both for 111 aircraft. The view average differs from a single figure for 248 aircraft, the ones drawn in two or three views.
- **Layer.** All six matrices pass the integrity checks. On the taxonomy criterion (same-G1 aircraft closer than different-G1 aircraft), **L24 cls** separates G1 best, averaged over the four rules. The experiment folders currently use L22 cls, so they are to be rebuilt on the winner before the probes.
- **Open.** Which rule and layer predict G1 best is decided by the probes (Section 3). The separation tests of Section 2 rank the options but do not train a classifier.

Section 1: Which Images Are Analysed

What this section establishes: which figures enter the embedding analysis, what each figure is labelled with, and the four rules that turn an aircraft's figures into the one vector a probe reads, with how much the rules disagree.

Figure selection: `eVTOL-Embedding-Extraction/notebooks/20_figure_selection` (gates) and `23_view_state_experiments` (rules, `src/view_state_experiments.py`). Rationale of each gate: `SELECTION_DECISIONS.md`.

1.1 From patents to whole-aircraft figures

Purpose: the gates every figure passes before any rule sees it.

The unit is the **aircraft** (`<patent>_ua<N>`): one patent can disclose several aircraft, and the G1 architecture and the main-figure marker are recorded per aircraft. Only approved figures of approved, in-domain aircraft are kept. The domain gate removes aircraft tagged UAV-similar, not electric, or STOL-only. D1 and D2 duplicates repeat an original and are removed, D3 variants stay. Only whole-aircraft figures are kept.

Table 1. Selection funnel (notebook 20).

Step	Figures removed	Figures remaining	Aircraft remaining
figures on file	0	5855	
figure not approved	3940	1915	841
image file missing	0	1915	841
patent not approved	2	1913	840
no aircraft row	0	1913	840
aircraft not approved	0	1913	840
domain gate: UAVSimilar	184	1729	754
domain gate: ElectricSimilar	66	1663	719
domain gate: STOLSimilar	3	1660	718
duplicate patent (D1/D2, points at its original)	74	1586	677
same image file as another figure row	1	1585	677
not a whole-aircraft figure	18	1567	677

The 2 aircraft marked unclassifiable in G1 have no class and are left out of every rule, which leaves **675 aircraft and 1563 figures**.

1.2 What each figure is labelled with

Purpose: the two figure labels the rules sort on, and how they are grouped.

View. The eight wizard views are grouped into four. "Generic 3D" is offered in the wizard but has not been used.

Table 2. Labelled views and their groups (whole-aircraft figures).

Labelled view (view8)	View group (view4)	Figures
Front-Isometric	Perspective	760
Rear-Isometric	Perspective	78
Top	Plan	314
Bottom/Down	Plan	10
Side	Side	261
Front	FrontRear	118
Back	FrontRear	26

Aircraft with at least one figure in each group: Perspective 510, Plan 271, Side 195, FrontRear 119.

Flight state. Only a convertible aircraft changes shape between hover and cruise, so only there does the flight state of a drawing matter.

- **Fixed types** (RC, MR, SLC, HB, PFV, TB, PTC): every figure is **Invariant** , whatever the label says.
- **Convertible types** (TW, TR, CVT, DS, SRW):
- **Hover** and **Cruise** are kept as labelled.
- A figure labelled Invariant, meaning the drawing fits both states, becomes **Both** .

- Transition and Other become `Other` , and an unlabelled figure is `Missing` .
- **SRW is convertible**, as in the wizard: its rotor stops and serves as the wing in cruise.

Table 3. Flight state of each figure, by G1 type.

G1	Group	Cruise	Both	Hover	Other	Missing	Invariant	Figures
TR	convertible	189	4	196	42	1	0	432
CVT	convertible	123	4	132	19	10	0	288
TW	convertible	71	6	77	12	2	0	168
SRW	convertible	9	4	4	1	0	0	18
DS	convertible	8	1	4	2	0	0	15
SLC	fixed	0	0	0	0	0	350	350
MR	fixed	0	0	0	0	0	107	107
TB	fixed	0	0	0	0	0	65	65
PTC	fixed	0	0	0	0	0	41	41
HB	fixed	0	0	0	0	0	34	34
PFV	fixed	0	0	0	0	0	23	23
RC	fixed	0	0	0	0	0	22	22

Canonical state: Cruise. Of 345 convertible aircraft, 215 have a Perspective figure in cruise and 209 one in hover. The margin is small. Cruise also suits the types whose defining feature only shows in forward flight, such as a stopped rotor used as a wing.

LABELS: VIEW AND STATE GROUPED

1.3 Four rules for one vector per aircraft

Purpose: how each rule chooses, so a difference in results can be traced to a difference in images.

Orderings. Each rule sorts the aircraft's whole-aircraft figures and takes the first.

- **View:** Perspective > Plan > Side > FrontRear.
- **State:** Cruise > Both > Hover > Other > Missing.
- **Ties:** the lowest FIG number wins, then the file name. Crops with no FIG label (`_Fu`) sort last.

Table 4. The four rules.

Rule	How the image is chosen	Vector
exp1 view-first	best view; within it, best state	that figure
exp2 state-first	best state; within it, best view	that figure
exp3 main	the figure the labeller marked as main	that figure
exp4 view average	best figure in each of Perspective, Plan, Side that the aircraft has	mean of 1 to 3 figure vectors, rescaled to length 1

exp1 and exp2 use the same two criteria in opposite order. When one figure is best on both, for example a Perspective drawing in cruise, both rules pick it. For fixed types the state never decides, so exp1 and exp2 always agree there. Two side tests place the slot vectors side by side instead of averaging them. They exist only for the aircraft that have every slot (Section 1.5).

Table 5. Fallbacks (the rule could not apply as written).

Rule	Aircraft	What was used
exp3	US2021309351A1_ua1	FALLBACK (main figure is not whole-vehicle): exp1's pick
exp3	US6367736B1_ua1	FALLBACK (main figure is not whole-vehicle): exp1's pick
exp4	ES2555162A1_ua1	FALLBACK (no Perspective/Plan/Side figure): slot FrontRear: state4=Invariant (rank 0), fig_number=?, of 1 figures

The two exp3 fallbacks have a part figure as their main and are fixed by marking a whole-aircraft main in the wizard.

1.4 How much the picks differ

Purpose: two rules can only score differently on the aircraft where they picked different images.

Table 6. Same figure picked by two single-figure rules.

Pair	Same figure (all)	Aircraft that differ	Same figure (convertible)	Convertible aircraft that differ
exp1 vs exp2	96.6 %	23	93.3 %	23
exp1 vs exp3	83.6 %	111	87.0 %	45
exp2 vs exp3	83.6 %	111	87.0 %	45

exp4 uses exactly exp1's figure for the 427 aircraft with a single slot and differs for the 248 drawn in two or three views.

Table 7. What each single-figure rule picks (views, then states).

Rule	Perspective	Plan	Side	FrontRear	state Invariant	state Cruise	state Both	state Hover	state Other	state Missing
exp1	508	139	27	1	330	247	6	61	23	8
exp2	494	136	35	10	330	269	5	43	20	8
exp3	485	124	39	27	330	249	6	57	25	8

Table 8. The 23 aircraft where exp1 and exp2 disagree.

exp1 picks	exp2 picks	Aircraft
Perspective + Hover	Side + Cruise	4
Perspective + Hover	Plan + Cruise	4
Plan + Hover	Side + Cruise	4
Perspective + Hover	FrontRear + Cruise	3
Plan + Hover	FrontRear + Cruise	3
Perspective + Both	FrontRear + Cruise	1
Perspective + Other	Plan + Cruise	1
Perspective + Other	FrontRear + Cruise	1
Plan + Other	Side + Hover	1
Side + Hover	FrontRear + Cruise	1

exp1 keeps the better view in a weaker state. exp2 moves to a weaker view to reach a better state, which is Cruise in 22 of the 23 cases.

Table 9. The 111 aircraft where the main figure differs from exp1: view of each pick.

exp3 (main) view	Perspective	Plan	Side
FrontRear	5	16	5
Perspective	48	0	0
Plan	11	5	0
Side	7	10	4

In 57 of the 111 the labeller's main has the same view as exp1's pick and is another drawing of it. In the rest the labeller preferred a different view.

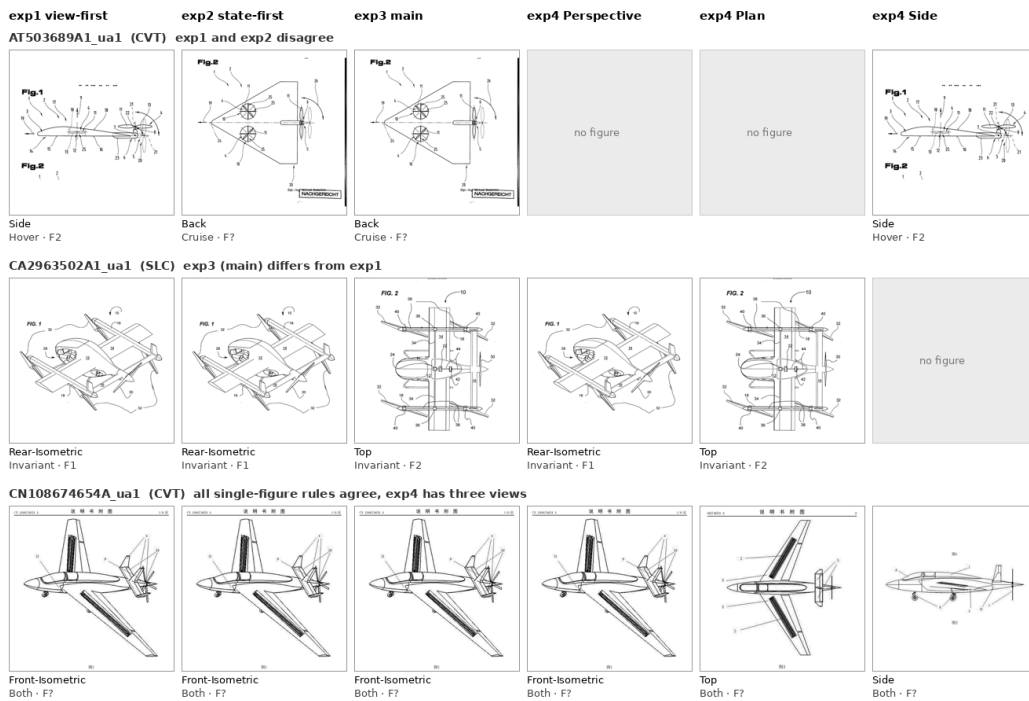


Figure 1. The images each rule uses for three aircraft (row 1, AT503689A1_ua1 (exp1 and exp2 disagree); row 2, CA2963502A1_ua1 (exp3 (main) differs from exp1); row 3, CN108674654A_ua1 (all single-figure rules agree, exp4 has three views)). Captions give the labelled view, the state group and the FIG number.

1.5 The view average (exp4)

Purpose: which aircraft the view average actually changes, and for which types it can show an effect.

A single drawing shows one view of the aircraft. The Plan view carries the wing and rotor layout, the Side view the vertical arrangement, the Perspective the overall shape. exp4 averages the views the aircraft has, on the assumption that the aircraft stays recognisable from one view to the next (tested in Section 2.4).

Table 10. Which slots the aircraft have.

Slots the aircraft has	Aircraft
Perspective	313
Plan	86
Perspective + Plan	81
Perspective + Side	64
Plan + Side	53
Perspective + Plan + Side	50
Side	27
FrontRear only	1

Table 11. Slots per aircraft, by G1 type.

G1	0 slots	1 slot	2 slots	3 slots	2 or 3 slots	Aircraft
SLC	0	123	48	14	62	185
TR	0	98	45	17	62	160
CVT	0	74	31	6	37	111
TW	0	30	28	2	30	60
MR	0	30	16	4	20	50
TB	0	14	10	4	14	28
PTC	1	21	3	1	4	26
HB	0	12	7	1	8	20
RC	0	8	2	1	3	11
PFV	0	7	3	0	3	10
SRW	0	4	4	0	4	8
DS	0	5	1	0	1	6

Types with few multi-view aircraft (see the "2 or 3 slots" column) cannot show an effect of the view average. The side tests keep the views apart: Perspective + Plan concatenated for the 131 aircraft that have both, and all three for the 50 that have every slot. They are compared with exp1 on the same aircraft only.

IMAGES: FOUR RULES DEFINED

Section 2: Which Layer

What this section tests: the six layer × pooling matrices on integrity, label-free structure, how well they keep same-G1 aircraft together under each rule, whether the drawing's view outweighs the architecture, and whether one aircraft stays recognisable across its views.

Matrices: `2_embedding_extraction/embeddings/dinov2-large_518/` (notebook 22, 2026-09-17, 1567 figures, vectors L2-normalised). Distances are cosine distances. Pairs from the same patent are left out of every separation test, so two drawings of one patent never count as a match. p-values from 999 label permutations (seed 42).

2.1 Integrity and structure against random noise

Purpose: that every matrix is valid and carries more structure than noise (label-free).

Integrity over all 1567 figures: no NaN, Inf, all-zero or duplicate vectors in any matrix.

Table 12. Label-free structure of each matrix (all figures).

Matrix	Cosine mean	Cosine SD	PC1 ÷ random	Effective dims	Dims for 90 %	Hopkins
L18 cls	0.969	0.016	51.3	16.1	72	0.757
L18 mean_patch	0.929	0.037	69.5	12.7	79	0.77
L22 cls	0.895	0.032	50.0	27.7	186	0.731
L22 mean_patch	0.928	0.033	58.3	14.6	94	0.757
L24 cls	0.407	0.118	37.2	40.9	272	0.736
L24 mean_patch	0.882	0.047	62.6	16.1	99	0.753

On its first principal component each matrix carries 37.2 to 69.5 times the variance of a random matrix of the same shape (pass bar 3). Hopkins above 0.5 means the vectors clump. These measures describe the matrices and do not use the taxonomy, so they do not decide the layer.

QC AND STRUCTURE: PASSED

2.2 Do same-G1 aircraft sit closer together?

Purpose: the taxonomy criterion, for every matrix under every rule.

Separation ratio = mean distance between aircraft of different G1 types ÷ mean distance between aircraft of the same type. A ratio of 1.0 means G1 is invisible to the matrix. **Cohen's d** is the same gap in standard deviations. 675 aircraft per cell.

Table 13. G1 separation: ratio (Cohen's d) per matrix and rule.

Matrix	exp1	exp2	exp3	exp4	Mean d
L18 cls	1.039 (d 0.08)	1.039 (d 0.08)	1.038 (d 0.07)	1.041 (d 0.08)	0.077
L18 mean_patch	1.047 (d 0.09)	1.049 (d 0.10)	1.047 (d 0.10)	1.055 (d 0.10)	0.098
L22 cls	1.045 (d 0.13)	1.043 (d 0.13)	1.046 (d 0.14)	1.045 (d 0.12)	0.129
L22 mean_patch	1.054 (d 0.12)	1.053 (d 0.12)	1.051 (d 0.11)	1.057 (d 0.12)	0.119
L24 cls	1.054 (d 0.25)	1.052 (d 0.24)	1.052 (d 0.24)	1.057 (d 0.23)	0.242
L24 mean_patch	1.052 (d 0.13)	1.051 (d 0.12)	1.054 (d 0.13)	1.059 (d 0.13)	0.129

Largest p-value in the table: 0.008. **L24 cls** has the highest mean effect over the four rules (0.242). Ranked by mean d over the six matrices, the rules come out as exp1 (0.134), exp3 (0.132), exp4 (0.132), exp2 (0.132). The rules differ by at most 0.002 in mean d, the matrices by 0.165. On this test the layer decides far more than the rule. This is a preview, not the verdict, since the probes measure prediction directly.

2.3 Viewpoint against architecture

Purpose: the July report found the drawing's viewpoint separating more strongly than any design attribute. Checked here on every matrix.

The same test on all figures, once with the view group as the label and once with G1. Pairs from the same patent are excluded, so the comparison is between different aircraft only.

Table 14. Separation by view group and by G1 (all figures, different patents only).

Matrix	View ratio	View d	G1 ratio	G1 d	View d ÷ G1 d
L18 cls	1.198	0.36	1.033	0.06	5.6
L18 mean_patch	1.219	0.38	1.046	0.09	4.42
L22 cls	1.204	0.63	1.036	0.12	5.39
L22 mean_patch	1.254	0.51	1.044	0.1	5.28
L24 cls	1.072	0.36	1.034	0.17	2.14
L24 mean_patch	1.2	0.47	1.041	0.1	4.63

A "View d ÷ G1 d" above 1 means two drawings in the same view look more alike than two aircraft of the same type. The lowest value is 2.14 (L24 cls) and the highest 5.6 (L18 cls). 6 of the 6 matrices weigh the view more than the architecture, so the view of the image a rule picks can move an aircraft further than its G1 type does.

2.4 Does an aircraft stay recognisable across its views?

Purpose: the assumption behind the view average (exp4).

Cosine similarity between the slot figures that exp4 averages. Two views of the same aircraft are compared with two different aircraft in the same view. The gap is expressed in standard deviations of the different-aircraft similarities, because the spread differs from layer to layer.

Table 15. Same aircraft across views against different aircraft.

Matrix	Same aircraft, other view	Other aircraft, same view	Other aircraft, other view	Gap (SD)	cos(exp4, exp1) multi-view
L18 cls	0.981	0.973	0.967	0.57	0.995
L18 mean_patch	0.944	0.94	0.924	0.13	0.985
L22 cls	0.922	0.906	0.887	0.49	0.98
L22 mean_patch	0.944	0.939	0.922	0.15	0.985
L24 cls	0.526	0.436	0.387	0.77	0.867
L24 mean_patch	0.912	0.897	0.875	0.33	0.976

A positive gap means that the same aircraft seen from another angle is closer than a different aircraft seen from the same angle. The largest gap is 0.77 SD (L24 cls). The last column shows how far averaging moves the vector away from exp1's single figure for the multi-view aircraft.

2.5 Layer declaration

L24 cls is carried forward. It separates G1 best averaged over the four rules (Table 13). On the other two criteria it stands as follows. View d ÷ G1 d is 2.14, against a range of 2.14 to 5.6 (Table 14). The across-view gap is 0.77 SD, against 0.13 to 0.77 (Table 15). Its label-free profile is PC1 ÷ random 37.2 and Hopkins 0.736.

The experiment folders were built on L22 cls (the July choice). Before the probes they are rebuilt on L24 cls, a change of one constant in notebook 23 (PRIMARY_LAYER , PRIMARY_POOLING). The choice is provisional: the probes of Section 3 run on every matrix and confirm or overturn it on prediction.

Section 3: Open Work

What remains: the tests that decide which rule, and which layer, gives the embedding that predicts the architecture best.

1. **Probes.** k-nearest-neighbour and logistic-regression probes predicting G1 from each rule's vectors on all six matrices, with folds grouped by patent and macro-F1 as the score. - exp4 is compared with exp1 on the 248 multi-view aircraft only, since the other 427 are identical in both. - The side tests are compared with exp1 on their own aircraft.
2. **Small classes.** RC 11, PFV 10, SRW 8, DS 6 have fewer than 20 aircraft. Their scores are reported with that caveat, or merged.
3. **Wizard fixes.** - Mark a whole-aircraft main for the two exp3 fallbacks. - Check whether KR20240170008A ua1 and ua2 are one aircraft drawn in two states.
4. **224 px.** The 224 px embeddings are not extracted yet. They are needed only for a resolution comparison.

SUMMARY: IN PROGRESS

Sources (read-only): `1639_LABELLED/2_embedding_extraction/selection/funnel.csv` (notebook 20); `.../embeddings/dinov2-large_518/` (notebook 22); `.../view_state_experiments/` (notebook 23: `figure_table`, `exp1-exp4 manifests`, `aircraft_common`, `exp_embeddings`). Every table is also written as CSV to `1639_LABELLED/3_embedding_evaluation/embedding_analysis/`. Labels: `@_labelling/outputs/tables/` (notebook 04).